Identity Sheet

Property	Value
Name	Urea
Other names	Carbamide, Carbonyl diamide, Diaminomethanal
Chemical formula	$CO(NH_2)_2$ — also written CH_4N_2O
Molecular weight	60.06 g/mol
CAS number	57-13-6
PubChem CID	1176
SMILES	NC(=O)N
Appearance	White, odourless, tasteless crystalline solid
Density	1.33 g/cm^3 (solid)
Melting point	132.7 °C (decomposes)

Structural formula:

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O
‖
H₂N—C—NH₂
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Urea is the simplest diamide of carbonic acid. It is a small, highly polar, uncharged molecule with strong hydrogen-bonding capacity.

Molecules Not to Be Confused With

Molecule	Formula	MW (g/mol)	Key Difference
Uric acid	$C_5H_4N_4O_3$	168.11	End-product of purine metabolism (not amino acid metabolism); poorly soluble; elevated in gout
Creatinine	$C_4H_7N_3O$	113.12	Breakdown product of creatine phosphate from muscle; used as GFR marker; different metabolic origin
Blood Urea Nitrogen (BUN)	—	—	Not a distinct molecule — it is the nitrogen component of urea. BUN = Urea × 28/60 (conversion factor 0.467). US labs report BUN in mg/dL; European labs report urea in mmol/L

Medical Information

Origin

Endogenous: Biosynthesis — The Urea Cycle Urea is the primary end-product of nitrogen metabolism in humans. It is synthesised almost exclusively in the **liver** (periportal hepatocytes) via the **urea cycle** (also called the ornithine cycle), first described by Krebs and Henseleit in 1932. The urea cycle converts toxic ammonia (NH_3), generated from amino acid deamination and other nitrogen sources, into the non-toxic, water-soluble urea molecule.

Steps of the urea cycle:

1. **Carbamoyl phosphate synthesis** (mitochondrial, rate-limiting): $NH_3 + CO_2 + 2\ ATP \rightarrow$ Carbamoyl phosphate. Catalysed by **carbamoyl phosphate synthetase I (CPS I)**, activated by N-acetylglutamate.
2. **Citrulline formation** (mitochondrial): Carbamoyl phosphate + Ornithine → Citrulline. Catalysed by **ornithine transcarbamylase (OTC)**.
3. **Argininosuccinate synthesis** (cytoplasmic): Citrulline + Aspartate + ATP → Argininosuccinate. Catalysed by **argininosuccinate synthetase (ASS)**.
4. **Arginine formation** (cytoplasmic): Argininosuccinate → Arginine + Fumarate. Catalysed by **argininosuccinate lyase (ASL)**.
5. **Urea release** (cytoplasmic): Arginine + H_2O → Urea + Ornithine. Catalysed by **arginase**. Ornithine is recycled back into the mitochondria.

Net reaction: $2\ NH_3 + CO_2 + 3\ ATP + H_2O \rightarrow$ Urea $+ 2\ ADP + AMP + 4\ P_i + PP_i$

A healthy adult produces approximately **10–20 g of urea per day** (≈ 170–330 mmol/day), depending primarily on dietary protein intake.

Exogenous Urea itself is not a significant dietary nutrient. However, its production is directly driven by dietary **protein intake**: amino acids from digested proteins are deaminated, and the resulting ammonia is converted to urea. Higher protein diets lead to higher urea production.

Urea is also present in some foods in trace amounts and is used as a food additive (E927b) in some countries. It is found in many topical skin products (5–40% concentration) as a moisturiser and keratolytic agent.

Primary & Secondary Biological Roles

Primary role: - **Nitrogen waste disposal**: Urea is the main vehicle for safe excretion of excess nitrogen. Approximately 80–90% of nitrogen excretion occurs as urinary urea.

Secondary roles: - **Renal concentrating mechanism**: Urea contributes significantly to the medullary osmotic gradient in the kidney via the countercurrent multiplication system. Urea transporters (UT-A1, UT-A2, UT-B) in the collecting duct and vasa recta allow urea recycling, which is critical for concentrating urine. - **Skin hydration**: Urea is a component of the Natural Moisturising Factor (NMF) in the stratum corneum, where it acts as a humectant. - **Protein denaturant**: At high concentrations (6–8 M), urea disrupts hydrogen bonds and hydrophobic interactions, denaturing proteins. This is exploited in biochemistry but has no known physiological role.

Catabolism and Elimination Pathway

- Urea is **freely filtered** at the glomerulus (100% filtered).
- Approximately **40–60% is passively reabsorbed** in the proximal tubule and thin descending limb.
- In the inner medullary collecting duct, ADH (vasopressin) regulates urea transporter UT-A1, permitting urea reabsorption into the medullary interstitium for the countercurrent concentrating mechanism.
- The remaining **40–60% is excreted in urine**.
- A small fraction (~15–40% of produced urea, typically 5–7 g/day) reaches the **gastrointestinal tract**, where it is hydrolysed by bacterial urease to ammonia and CO_2. The ammonia is either re-absorbed and re-converted to urea (enterohepatic nitrogen cycle) or excreted in faeces.
- Minor elimination occurs via **sweat** (~1–2 g/day under normal conditions, higher with profuse sweating).

Expression in Humans

Normal Levels

Compartment	Reference Range
Serum/Plasma urea	2.5–7.1 mmol/L (15–43 mg/dL) in adults
BUN	7–20 mg/dL (2.5–7.1 mmol/L) in adults 18–59 y
BUN (>60 y)	8–23 mg/dL
BUN (children)	5–18 mg/dL
BUN (pregnancy)	5–12 mg/dL (lower due to haemodilution and increased GFR)
Urinary urea	12–20 g/24 h (200–333 mmol/24 h)
Urinary urea concentration	200–400 mmol/L (highly variable with hydration)

Factors Influencing Levels **Increased urea:** - High protein diet - Gastrointestinal bleeding (digestion of blood proteins) - Catabolic states (fever, burns, trauma, sepsis, corticosteroid therapy) - Dehydration / hypovolaemia (prerenal azotemia) - Renal impairment (reduced GFR) - Post-renal obstruction - Tetracycline (increases protein catabolism)

Decreased urea: - Low protein diet or malnutrition - Severe liver disease (impaired urea cycle) - Overhydration / SIADH - Pregnancy (haemodilution + increased GFR) - Anabolic states (growth hormone) - Inborn errors of the urea cycle

Associated Pathologies

Condition	Urea/BUN Level	Key Symptoms
Azotemia (prerenal)	Elevated BUN, BUN/Creatinine ratio >20:1	Dehydration, hypotension, oliguria
Azotemia (renal / intrinsic)	Elevated BUN, BUN/Creatinine ratio ~10-15:1	Oedema, hypertension, haematuria
Azotemia (post-renal)	Elevated BUN	Obstruction symptoms, anuria
Uremia	Markedly elevated (typically BUN >60 mg/dL)	Nausea, vomiting, pericarditis, encephalopathy, asterixis, uremic frost
Chronic kidney disease (CKD)	Progressive elevation	Fatigue, anaemia, bone disease, electrolyte disturbances
Acute kidney injury (AKI)	Rapidly rising	Oliguria, fluid overload
Liver failure / cirrhosis	Low urea	Jaundice, coagulopathy, hepatic encephalopathy (from accumulating ammonia)
Urea cycle disorders	Low urea, hyperammonaemia	Neonatal encephalopathy, vomiting, seizures, intellectual disability
GI bleeding	Elevated BUN, BUN/Creatinine >20:1	Melena, haematemesis

Presence in Urine

Should it be normally present? Yes — urea is the most abundant organic solute in human urine, constituting approximately **25–50% of all urinary solids** by weight.

Normal urinary levels: - 24-hour excretion: **12–20 g/day** (200–333 mmol/day) - Urine concentration: highly variable, typically **200–400 mmol/L** but can range from <100 mmol/L (dilute) to >700 mmol/L (concentrated) - Urine Urea Nitrogen (UUN): **6–17 g/day** (equivalent to urinary urea × 0.467)

Form in urine: Urea is present in its **native, dissolved molecular form** — as an intact, uncharged molecule ($CO(NH_2)_2$). It does not ionise under physiological urinary pH conditions (pKa ≈ 0.1 for protonation of carbonyl oxygen, effectively non-ionisable in the pH 4.5–8.0 range of urine).

Pathological significance of abnormal urinary urea:

Urinary Urea	Possible Causes	Prevalence
Elevated	High protein diet, catabolic states, fever, hyperthyroidism	Common
Decreased	Liver failure, low protein intake, renal failure (late stage with reduced GFR), urea cycle defects	Urea cycle disorders: ~1:35,000 births

The urea-to-creatinine ratio in urine can help assess protein intake adequacy and nitrogen balance status.

Solubility: - **In water:** ~1080 g/L at 20 °C (extremely soluble) - Solubility increases with temperature: 1670 g/L (40 °C), 2510 g/L (60 °C), 7330 g/L (100 °C) - Urea is never a cause of urinary crystalluria or stone formation due to its extraordinary solubility.

Detection in Urine

Available Clinical Assays

1. **Enzymatic urease–GLDH coupled assay** (most widely used automated method):
   - **Principle:** Urease hydrolyses urea → 2 NH_4^+ + CO_2. Then glutamate dehydrogenase (GLDH) catalyses: NH_4^+ + α-ketoglutarate + NADH → L-glutamate + NAD^+ + H_2O.
   - **Detection:** Decrease in NADH absorbance at **340 nm** is proportional to urea concentration.
   - **Advantages:** Specific, automatable, fast (kinetic assay), minimal interference.
2. **Diacetyl monoxime (DAM) colorimetric method:**
   - **Principle:** Urea reacts with diacetyl (generated from diacetyl monoxime under acidic conditions and heat) to form a pink-red diazine chromophore.
   - **Detection:** Absorbance at **540 nm** (some protocols cite 515–540 nm).
   - **Advantages:** Simple, low-cost, suitable for manual and semi-automated settings.
   - **Disadvantages:** Requires strong acids and heating; less specific than enzymatic methods.
3. **Conductimetric method:** Measures conductivity change as urease converts uncharged urea to ionic ammonium and bicarbonate. Less common in routine labs.
4. **Berthelot (indophenol) method:** Ammonia from urease reaction is converted to indophenol (blue dye) using phenol and hypochlorite. Measured at ~630 nm.

Optimal Urine Type for Measurement

- The **24-hour urine collection** is the gold standard for quantifying total daily urea excretion, essential for nitrogen balance studies and assessment of protein intake.
- **Spot urine** can be used with urea/creatinine ratio normalisation for quick assessments.
- Urea levels vary throughout the day with protein intake and hydration. First morning urine tends to be more concentrated, giving higher absolute urea concentrations, but this does not provide better diagnostic accuracy than a properly collected 24-hour specimen.

- **Recommendation:** 24-hour collection for quantitative assessment; spot urine with creatinine correction for screening.
- Urine should be preserved at **2–8 °C** or with thymol/boric acid to prevent bacterial urease activity from artificially lowering urea and increasing ammonia.

Actual Gold Standard

The **enzymatic urease–GLDH kinetic UV assay** is the current gold standard for both blood and urine urea measurement. It is used on all major automated chemistry platforms (Roche Cobas, Siemens Advia, Abbott Architect, Beckman AU). It measures NADH consumption at 340 nm in a kinetic mode, providing excellent specificity and precision (CV <3%).

Interferences in Measurement

Interference	Effect	Mechanism
Ammonia contamination	False elevation	Ammonia is the measured intermediate; environmental or bacterial ammonia raises apparent urea
Bacterial contamination	False decrease	Bacterial urease degrades urea before analysis
Fluoride (NaF)	False decrease	Inhibits urease enzyme in the assay
Haemolysis	Variable	Releases intracellular ammonium; may interfere with NADH absorbance at 340 nm
Lipaemia	Positive bias	Light scattering affects absorbance measurement
High bilirubin	Negative bias	Absorbs near 340 nm in some assay configurations
Chloramphenicol, streptomycin	May inhibit urease	Drug interference at very high concentrations
Ascorbic acid	Minor interference	Competes in some colorimetric assays
DAM method-specific: thiourea, guanidine	False elevation	React with diacetyl to form similar chromophores

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- **UV-Vis:** Urea has no significant UV-Vis chromophore; it does not absorb appreciably in the 200–800 nm range (very weak absorption <200 nm). Direct UV-Vis detection is therefore **not feasible**. All clinical assays rely on coupled reactions to generate a measurable chromophore or consume a UV-active cofactor (NADH at 340 nm).
- **NIR spectroscopy:** Urea absorbs in the near-infrared via overtone and combination bands of N–H and C=O stretches:
  - **~2150 nm** (combination band of NH_2 deformation + C=O stretch)
  - **~1500 nm** (first overtone of N–H stretch)
  - NIR has been used for non-invasive blood urea monitoring and urine analysis with multivariate calibration (PLS regression).

- **LOD:** ~1–5 mM in aqueous solution (matrix-dependent); adequate for urinary urea concentrations (200–400 mM).
- **Sample prep:** None required (direct measurement of liquid urine); some systems use fibre-optic probes.

Fluorescence Detection Urea itself is **non-fluorescent**. All fluorescence-based approaches use indirect detection via enzyme-coupled or chemosensor systems:

1. **Urease + pH-sensitive fluorophores:**
   - Urease hydrolyses urea → NH_3 + CO_2, raising local pH.
   - pH-sensitive fluorescent probes (e.g., fluorescein derivatives, SNARF-1) report the pH change.
   - **Excitation/Emission:** depends on the chosen fluorophore (e.g., fluorescein: Ex 490 nm / Em 520 nm).
   - **LOD:** 0.1–1 mM (sufficient for clinical urine levels).
2. **Quantum dot (QD) systems:**
   - **CdSe/ZnS QDs + urease:** pH increase from urea hydrolysis quenches or enhances QD fluorescence. LOD: ~0.5 mM. Ex ~350–400 nm / Em ~520–580 nm.
   - **ZnS QDs capped with glutathione + urease:** Non-enzymatic detection by PL quenching; LOD 0.5–5 mM range. Ex 320 nm / Em 430 nm.
   - **$CuInS_2$ QDs functionalised with dopamine:** pH-responsive fluorescence; LOD ~0.1 mM.
   - **Carbon nanodots (N-CDs):** pH-dependent blue fluorescence; Ex 360 nm / Em 440 nm; LOD ~0.05 mM in buffer.
3. **Ratiometric fluorescence sensors:**
   - Dual-emission systems (e.g., R-CDs/methyl red + NIR-CDs/Cu^{2+}) where urea hydrolysis-driven pH change modulates two emission channels differently; LOD ~0.01–0.1 mM.

Sample pretreatment: Most fluorescence sensors require urease addition and controlled pH buffering. Urine samples may require dilution to reduce matrix quenching effects. Centrifugation recommended for turbid samples.

Raman Detection Urea has **characteristic Raman peaks** allowing direct detection:

Peak (cm^{-1})	Assignment
1003	C–N symmetric stretch (νs)
1540	NH_2 bending (δ)
~1640	C=O stretch (ν)
~3340–3440	N–H stretches

- **Conventional Raman:** Can detect urea in urine, but sensitivity is limited. Urea has a relatively low Raman scattering cross-section. LOD ~5–10 mM in aqueous solutions. Adequate for clinical urine concentrations but marginal.
- **Surface-Enhanced Raman Spectroscopy (SERS):**
  - **Fe_3O_4@C@Ag nanostructures:** LOD = 5.68×10^{-9} M (5.68 nM), enhancement factor 3.67×10^6. Validated in spiked urine with 90–99% recovery (Unlocking new possibilities, RSC Advances, 2025).
  - **Gold nanoparticle@Al foil substrates:** Cost-effective SERS for urea in artificial urine (Analytical Sciences, 2018).
  - **Ag/Au nanoparticle colloids:** Used for urinary metabolite profiling including urea.
  - **Excitation wavelengths:** Typically 532 nm, 633 nm, or 785 nm laser lines.
  - **Sample prep:** For SERS, urine samples are typically mixed with colloidal nanoparticles or deposited on SERS substrates. Some protocols require dilution or salt addition for nanoparticle aggregation.

FTIR Detection Urea has strong, well-characterised **infrared absorption bands:**

Band (cm^{-1})	Assignment	Notes
~1627–1640	C=O stretching (ν)	Overlaps with water bending mode (~1640 cm^{-1})
~1585–1600	NH_2 bending (δ)	Asymmetric deformation
~1460–1468	C–N antisymmetric stretching (νas)	**Most diagnostic band for quantification**; shifts with hydration state (1468 cm^{-1} in solution → 1443–1464 cm^{-1} when dried)
~3340, 3440	N–H stretching	Symmetric and antisymmetric

- **ATR-FTIR** (Attenuated Total Reflection) is the preferred modality for urine analysis.
- **Quantification:** The νas(CN) band at ~1468 cm^{-1} (in solution) is used for quantification. Shaw et al. (2000) demonstrated reagent-free quantification of urea, creatinine, and total protein in dried urine films using mid-IR spectroscopy.
- **Hydration effects:** Rich & Iwaki (2017, Applied Spectroscopy, PMC5379246) demonstrated that urea in dried urine adopts intermediate hydration states (1443–1454 cm^{-1}) rather than the fully dry form (1464 cm^{-1}), complicating spectral decomposition. Four components are needed for accurate fitting.
- **LOD:** Not a limiting factor — urea is the dominant organic solute in urine, producing strong IR bands well above detection limits.
- **Sample prep:** Liquid urine can be measured directly on an ATR crystal; dried films increase band intensities but introduce hydration-state artefacts. No chemical pretreatment needed.
- **Interferences:** Water O–H bending mode at 1640 cm^{-1} overlaps with urea C=O stretch. ATR mode with water subtraction algorithms addresses this. Creatinine bands overlap slightly at ~1492 cm^{-1}.

Voltammetry Detection Direct electrochemical oxidation of urea is difficult because it requires a high overpotential. Two main approaches exist:

1. **Enzyme-based (urease) electrochemical biosensors:**
   - Urease immobilised on electrode surface; urea hydrolysis produces NH_4^+ and HCO_3^-.
   - Detection via amperometric (NH_4^+ oxidation), potentiometric (pH/NH_4^+ ion-selective electrode), or conductimetric transduction.
   - **LOD:** 10–100 μM range, depending on electrode design.
   - **Linear range:** Typically 0.1–100 mM; well-suited for urinary concentrations.
   - **Sample prep:** Urine may need dilution and pH adjustment. Urease stability and reusability are key challenges.
2. **Non-enzymatic (catalyst-based) sensors:**
   - **Nickel-based electrodes (NiO, NiOOH, Ni nanoparticles):** In alkaline media (0.1–1 M NaOH), Ni^{2+}/Ni^{3+} redox couple catalyses urea oxidation.
     - NiO nanostructures on GCE: LOD = 10 μM, linear range 100–1100 μM (Arain et al., RSC Advances, 2016).
     - Ni nanoparticles on GCE: selective urea sensing in complex matrices (Scientific Reports, 2025).
     - $Ni(bzimpy)_2Cl_2$ complex: LOD = 1.5 μM, linear range 10–100 μM (Electrochimica Acta, 2021).
   - **Techniques used:** Cyclic voltammetry (CV), linear sweep voltammetry (LSV), chronoamperometry, differential pulse voltammetry (DPV).
   - **Limitation:** Alkaline media requirement; urine pH must be adjusted. Selectivity against other electroactive species (ascorbic acid, uric acid, creatinine).

Other Detection Technologies

1. **Colorimetric paper-based / lateral flow assays:**
   - Urease immobilised on paper substrate; pH indicator (bromothymol blue, phenol red) changes colour upon urea hydrolysis.
   - Smartphone-based readout for semi-quantitative results.
   - **LOD:** ~1–5 mM; suitable for screening but not precise quantification.
   - **Advantage:** Low cost (<$0.50/test), no instrumentation, point-of-care.
2. **Microfluidic / lab-on-chip:**
   - Integrated urease reactors with electrochemical or optical readout.
   - Enables automated sample handling with μL volumes.
   - **LOD:** ~0.05–1 mM depending on detection modality.
3. **Molecularly imprinted polymers (MIPs):**
   - Synthetic recognition elements mimicking antibody-like binding to urea.
   - Combined with electrochemical or colorimetric transduction.
   - **Advantage:** No enzyme needed; robust and reusable.
   - **Limitation:** Selectivity can be lower than enzymatic approaches for the small, simple urea molecule.
4. **Conductimetric sensors:**
   - Urease-based sensors measuring conductivity increase (uncharged urea → ionic products).
   - Simple instrumentation; LOD ~0.01–0.1 mM.
   - Used in some commercial biosensors.
5. **Thermal / calorimetric biosensors (enzyme thermistors):**
   - Measure the exothermic heat of urea hydrolysis by urease.
   - **LOD:** ~0.05 mM.

Summary Table: Detection Methods Comparison

Method	LOD	Wavelength / Key Parameter	Sample Prep	Strengths	Limitations
Urease–GLDH (gold standard)	~0.1 mM	340 nm (NADH)	Dilution for urine	Automated, precise, specific	Reagent cost, instrument needed
DAM colorimetric	~0.5 mM	540 nm	Acid + heat	Simple, low cost	Less specific, toxic reagents
NIR spectroscopy	~1–5 mM	1500 nm, 2150 nm	None	Reagent-free, non-destructive	Requires chemometric calibration
ATR-FTIR	N/A (urea dominant in urine)	1468 cm^{-1} (νas CN)	None or drying	Reagent-free, multianalyte	Instrumentation cost, water overlap
Raman (conventional)	~5–10 mM	1003 cm^{-1} (C–N), excitation 532/785 nm	None	Non-destructive	Low sensitivity

Method	LOD	Wavelength / Key Parameter	Sample Prep	Strengths	Limitations
SERS	~5.7 nM (best)	1003 cm^{-1}, excitation 532/633/785 nm	Nanoparticle mixing	Ultra-sensitive	Substrate cost, reproducibility
Fluorescence (QD/enzyme)	0.01–1 mM	Ex 320–490 nm / Em 430–580 nm	Urease + buffer	Sensitive, versatile	Indirect; enzyme stability
Ni-electrode voltammetry	1.5–10 μM	Oxidation peak ~0.5 V vs Ag/AgCl	Alkaline pH adjustment	Enzyme-free, low cost	Requires alkaline media
Urease amperometric	10–100 μM	Applied potential varies	pH buffer	Specific	Enzyme degradation
Paper-based colorimetric	~1–5 mM	Visual / smartphone	None	Ultra low cost, portable	Semi-quantitative only

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